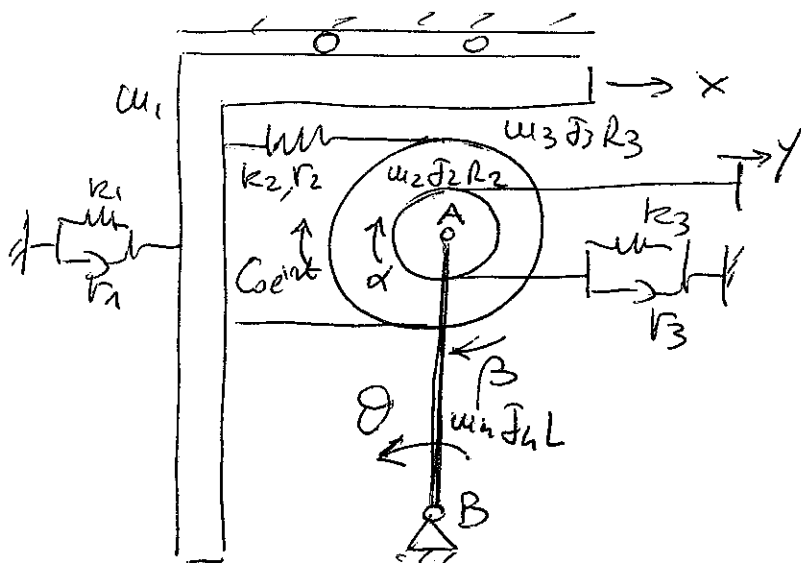




$$\underline{x}_E = \begin{Bmatrix} x \\ \theta \end{Bmatrix}$$

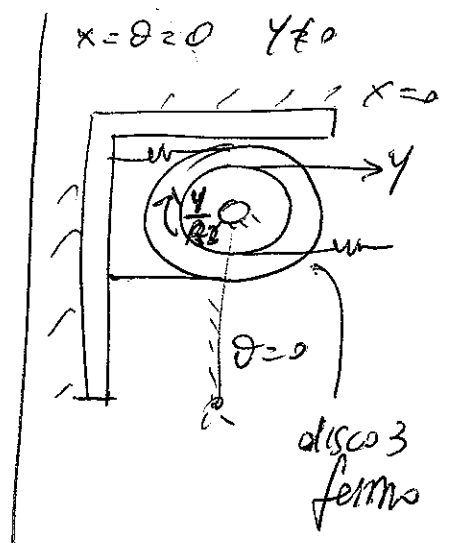
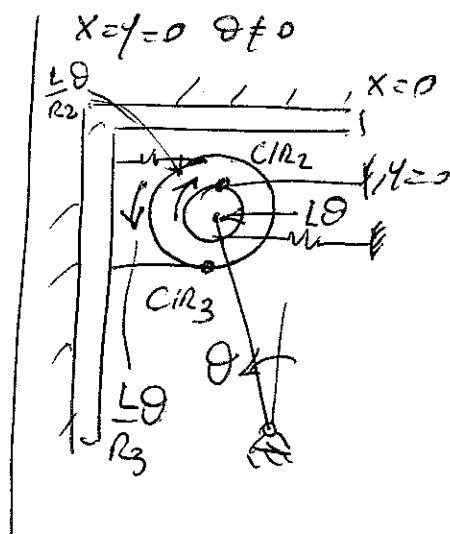
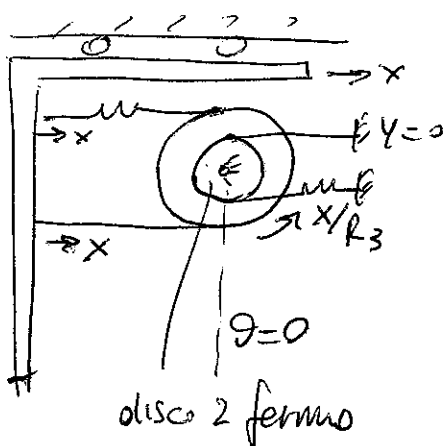
$$\underline{x} = \begin{Bmatrix} x \\ \theta \\ y \end{Bmatrix}$$

$$T = \frac{1}{2} m_1 \dot{x}^2 + \frac{1}{2} (m_2 + m_3) \dot{x}_A^2 + \frac{1}{2} J_2 \dot{\alpha}^2 + \frac{1}{2} J_3 \dot{\beta}^2 + \frac{1}{2} m_4 \dot{x}_s^2 + \frac{1}{2} J_4 \dot{\theta}^2$$

$$\underline{y}_{ME} = \begin{Bmatrix} x & x_A & \alpha & \beta & x_s & \theta \end{Bmatrix}^T \quad [m_E] = \text{diag}(m_1, m_2 + m_3, J_2, J_3, m_4 J_4)$$

$$\begin{cases} x_A = -L\theta \\ \alpha = \frac{L}{R_2} \theta + \frac{y}{R_2} \\ \beta = -\frac{x}{R_3} - \frac{L}{R_3} \theta \\ x_s = -\frac{L}{2} \theta \end{cases} \quad [A_{ME}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -L & 0 \\ 0 & L/R_2 & 1/R_2 \\ -1/R_3 & -L/R_3 & 0 \\ 0 & -L/2 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$x \neq 0 \quad \theta = y = 0 :$



$$V_{EL} = \frac{1}{2} k_1 \Delta L_1^2 + \frac{1}{2} k_2 \Delta L_2^2 + \frac{1}{2} k_3 \Delta L_3^2$$

$$\underline{q}_k = \{ \Delta L_1 \quad \Delta L_2 \quad \Delta L_3 \}^T \quad (k_F = \text{diag}(k_1 \quad k_2 \quad k_3))$$

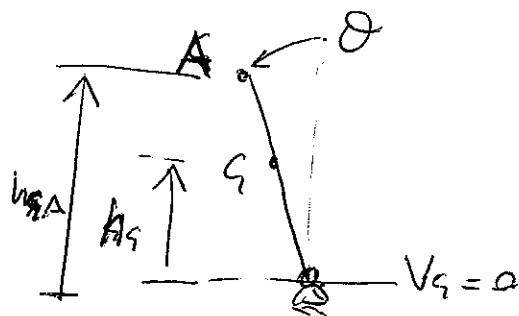
$$\begin{cases} \Delta L_1 = x \\ \Delta L_2 = -2x - 2L\theta \\ \Delta L_3 = 2L\theta + y \end{cases} \quad [A_k] = \begin{bmatrix} 1 & 0 & 0 \\ -2 & -2L & 0 \\ 0 & 2L & 1 \end{bmatrix}$$

molle e smorzatori in parallelo + dinamica

linearizzato $\rightarrow [A_r] = [A_k]; \quad r_F = \text{diag}(r_1 \quad r_2 \quad r_3)$

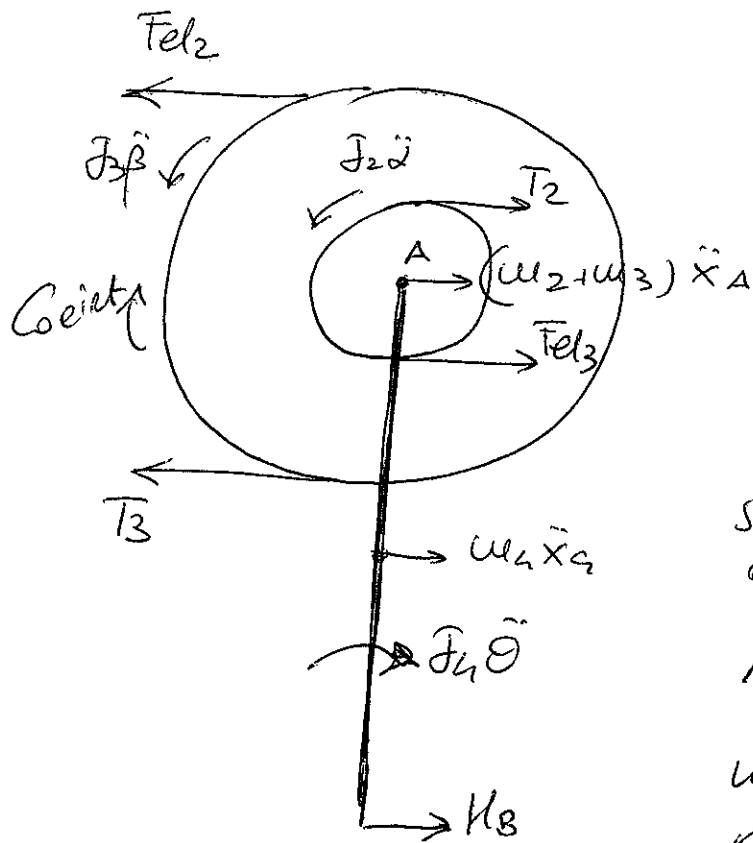
$$V_g = (m_2 + m_3) g h_{gA} + m_4 g h_g$$

$$\begin{cases} h_{gA} = L \cos \theta \\ h_g = \frac{L}{2} \cos \theta \end{cases}$$



$$\rightarrow [k_g] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -(m_2 + m_3 + \frac{m_4}{2}) g L & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\delta L = C_0 e^{i\omega t} \left(-\frac{\delta x}{R_3} - \frac{L}{R_3} \delta \theta \right) \rightarrow \underline{Q} = \begin{bmatrix} -1/R_3 \\ -L/R_3 \\ 0 \end{bmatrix} C_0 e^{i\omega t}$$



scrivere solo le forze
orientate e le
coppie.
Poi scrivere le forze
verticali, ma
non vengono usate
nelle equazioni!

$$\sum \vec{F}_X^{ASTA + DISCO2 + DISCO3} = 0 \quad : \quad H_B + u_4 \ddot{x}_4 + (\omega_2 + \omega_3) \ddot{x}_A - T_3 - F_{el2} + T_2 + F_{el3} = 0 \quad (1)$$

$$\begin{cases} F_{el2} = (k_2 + i \omega r_2) \Delta l_2 \\ F_{el3} = (k_3 + i \omega r_3) \Delta l_3 \end{cases}$$

$$\sum M_A^{DISCO2} = 0 \quad : \quad T_2 R_2 - J_2 \ddot{\alpha} - F_{el3} R_2 = 0 \rightarrow T_2 = F_{el3} + \frac{J_2 \ddot{\alpha}}{R_2}$$

$$\sum M_A^{DISCO3} = 0 \quad : \quad T_3 R_3 - J_3 \ddot{\beta} - F_{el2} R_3 + C_{eint} = 0 \rightarrow$$

$$\rightarrow T_3 = F_{el2} + \frac{J_3 \ddot{\beta}}{R_3} - \frac{C_{eint}}{R_3}$$

sostituire nelle (1) $\rightarrow H_B = \dots$